

Luca Calculations Audit Packet

Compiled audit packet

This packet combines the two complementary audit views.

1. **In-place annotated original.** This is the author-facing redline version. Luca's original manuscript order is preserved, wrong local steps remain visible, and the corrected formulae are inserted immediately after the problematic steps.
2. **Complete corrected derivation.** This is the expanded derivation behind the redline replacements. It is intended as the no-skipped-steps companion proof for checking the algebra and the corrected tail conclusion.

Calculations for network paper

(Dated: June 1, 2026)

AUDIT NOTE: how this annotated copy should be read

This file preserves Luca's original manuscript text and equation order as far as possible. The inserted boxes are the audit layer. Original equations are left in place, even when they are not valid under the stochastic interpretation used here. The correction starts from the model in which a shortcut from u into the absorbing site v removes probability from the transient state space. Under that interpretation the shortcut is a killing channel at u , not a probability-conserving transfer to another transient site.

Source of truth: `source/original_from_outlook_Calculations.tex`. The original source file is archived separately and is not modified by this annotated copy.

Companion full derivation: `Calculations_corrected_complete_v3.tex` contains the expanded 21-page corrected derivation. The present file is the in-place redline map; the companion file is the no-shortcut, no-redline derivation trail behind the blue replacement formulae.

PRELIMINARIES

These are preliminaries identities that can be derived by using the finite series identities in PRX. I have derived the following for $|m| < N$ and $m \neq 0$

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} = 1 \quad (1)$$

Another identity exploits the fact that

$$\begin{aligned} [1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] &= \frac{1}{2} \left\{ \cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right] \right. \\ &\quad \left. - \frac{1}{2} \left[\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right] \right] \right\} \end{aligned}$$

for a generic σ we have

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} &= \frac{1}{2} \sum_{k=1}^N \frac{\cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &\quad - \frac{1}{4} \sum_{k=1}^{N-1} \frac{\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &= \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \end{aligned} \quad (2)$$

If $\sigma = \cos \left[\frac{2\pi r}{N} \right]$ one needs to use de l'Hospital rule to evaluate the expression above, which gives

$$\begin{aligned} \lim_{\sigma \rightarrow \cos \left[\frac{2\pi r}{N} \right]} \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \\ = \cos \left[\frac{2\pi r|m|}{N} \right], \end{aligned} \quad (3)$$

in other words we have the general identity for $0 < |m| < N$ and $r = 1, \dots, N - 1$,

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\cos \left[\frac{2\pi r}{N} \right] - \cos \left[\frac{\pi k}{N} \right]} = \cos \left[\frac{2\pi r|m|}{N} \right], \quad (4)$$

The third relevant identity for $|m| \leq N$ and integer k is given by

$$\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{(2k-1)\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = \frac{1}{N \left\{ 1 - \cos \left[\frac{(2k-1)\pi}{N} \right] \right\}} - \frac{\sin \left[\frac{(2k-1)\pi |m|}{N} \right]}{\sin \left[\frac{(2k-1)\pi}{N} \right]} \quad (5)$$

alternatively

$$\frac{1}{N} \sum_{r=1}^{N-1} [1 - (-1)^k] \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{k\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = [1 - (-1)^k] \left\{ \frac{1}{N \left\{ 1 - \cos \left[\frac{\pi k}{N} \right] \right\}} - \frac{\sin \left[\frac{\pi k |m|}{N} \right]}{\sin \left[\frac{\pi k}{N} \right]} \right\} \quad (6)$$

PROPAGATOR IN RING LATTICE WITH ABSORPTION SITE

The occupation probability for the periodic ring lattice with arbitrary K and an absorption site at $n = m$ can be written via the general relation for any $t \geq 1$

$$W_{n_0}(n, t) = Q_{n_0}^{(k)}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0 \rightarrow m}(s) Q_m^{(k)}(n, t-s) \quad (7)$$

where $Q_r^{(k)}(n, t)$ is the propagator for the periodic ring lattice in the absence of any absorption site. Note that at $t = 0$, by construction $F_{n_0 \rightarrow m}(0) = 0$ when $n_0 \neq n$, and we have $W_{n_0}(n, 0) = Q_{n_0}^{(k)}(n, 0)$.

More explicit expressions can be obtained for the case $K = 2$, that is the nearest-neighbour lattice walk. We have that

$$Q_{n_0}^{(1)}(n, t) = \frac{1}{N} \sum_{k=0}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t, \quad (8)$$

and for $t \geq 1$

$$\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{|m-n_0|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^{t-1}. \quad (9)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$.

Given that in Eq. (9) only odd k terms are non-zero, the first-passage probability can be written in compact form as

$$\mathcal{F}_{n_0}(m, t) = \sum_{k=1}^{\frac{N}{2}} f_k(n_0, m) \alpha_k^{t-1}. \quad (10)$$

where

$$f_k(n_0, m) = \frac{2q}{N} \sin \left[\frac{|m-n_0|\pi(2k-1)}{N} \right] \sin \left[\frac{\pi(2k-1)}{N} \right] \quad (11)$$

and

$$\alpha_k = 1 - q + q \cos \left[\frac{\pi(2k-1)}{N} \right]. \quad (12)$$

The generating function of Eq. (10) in compact form can be written as

$$\tilde{\mathcal{F}}_{n_0}(m, z) = z \sum_{k=1}^{N-1} \frac{f_k(n_0, m)}{1 - z\alpha_k} = \frac{U_{N-|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + U_{|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (13)$$

where $U_r(y)$ is the Chebyshev polynomial of the second kind.

Also we note that using Eq. (1) one can show that the survival probability

$$\sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = 1 - \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t,$$

which provides the survival probability

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t.$$

Using the above expressions in Eq. (7) and evaluating the time summation, we obtain

$$\begin{aligned} W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n - n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\ &+ \frac{1}{N^2} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\ &+ \frac{1}{N^2} \sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \\ &\times \left\{ \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \right\} \end{aligned} \quad (14)$$

Using the identities (4) and (6), we reduce the double summation to single summations as follows

$$\begin{aligned} &\sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \left\{ \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\} \\ &= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0 - m|}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\ &- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} - N \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{n - m| \pi k}{N} \right] \right\} \\ &\times \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\ &= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0 - m|}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\ &- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \right. \\ &\left. - N \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{n - m| \pi k}{N} \right] \right\} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \end{aligned} \quad (15)$$

and we have

$$\begin{aligned}
W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&\quad - \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&\quad + \frac{1}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] \left\{ 1 - q + q \cos \left[\frac{\pi k}{N} \right] \right\}^t.
\end{aligned} \tag{16}$$

As $\sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] = -1$ and $\sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] = 0$, one can show that $Q_{n_0}(n, 0) = \delta_{n, n_0}$. Furthermore when either $n = m$ or $n_0 = m$, one has that $Q_{n_0}(n, t) = 0$ for all t , and it is also easy to see that $Q_{n_0}(n, t \rightarrow \infty) = 0$, which means that all random walks eventually reach the absorbing site $n = m$.

We can rewrite Eq. (16) as

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t + G_k^{(2)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \tag{17}$$

where m is the absorbing site, and with

$$\begin{aligned}
G_k^{(1)}(n_0, n, m) &= \frac{\cos \left[\frac{2\pi k(n-n_0)}{N} \right] - \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right]}{N} \\
G_k^{(2)}(n_0, n, m) &= \frac{[1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right]}{N}
\end{aligned} \tag{18}$$

In terms of generating functions one has

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]} + \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]} \tag{19}$$

or the equivalent form

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z\gamma_k} + \sum_{k=1}^{\frac{N}{2}} \frac{g_k^{(2)}(n_0, n, m)}{1 - z\alpha_k}, \tag{20}$$

and its time-dependent version

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \gamma_k^t + \sum_{k=1}^{\frac{N}{2}} g_k^{(2)}(n_0, n, m) \alpha_k^t, \tag{21}$$

where α_k is defined in Eq. (12),

$$\gamma_k = 1 - q + q \cos \left(\frac{2\pi k}{N} \right), \tag{22}$$

and

$$g_k^{(2)}(n_0, n, m) = \frac{2}{N} \sin \left[\frac{\pi(2k-1)|n_0-m|}{N} \right] \sin \left[\frac{\pi(2k-1)|n-m|}{N} \right]. \tag{23}$$

The associated expressions in terms of Chebyshev polynomials of the first and second kind, respectively, $T_r(y)$ and

$U_r(y)$, are

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \left[1 - q + q \cos\left(\frac{2\pi k}{N}\right) \right]} &= \left\{ 2T_{N-|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + 2T_{|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &- T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \\ &- T_{N-|n_0-m|+|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|+|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \Big\} \\ &\times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (24)$$

and

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos\left(\frac{\pi k}{N}\right) \right]} &= \left\{ T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &- T_{|N-n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \Big\} \\ &\times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (25)$$

When $n = n_0$ Eq. (17) reduces to

$$\begin{aligned} W_n(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \left\{ \left(1 - \cos^2 \left[\frac{2\pi k(n-m)}{N} \right] \right) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \right. \\ &\quad \left. + [1 - (-1)^k] \sin^2 \left[\frac{\pi k|n-m|}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\}, \end{aligned} \quad (26)$$

whose generating function is given by the compact expression

$$\begin{aligned} \widetilde{W}_n(n, z) &= \left\{ 2T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &- T_{N-|2|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|2|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \Big\} \\ &\left\{ 2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\}^{-1} \end{aligned} \quad (27)$$

Note that using Eqs. (19) with $n = n_0$ and (27) when $|n-m| = N/2$ with N even, one can show setting $z = 1$ the following identities

$$\sum_{k=1}^{N-1} \frac{1 - (-1)^k}{1 - \cos\left(\frac{\pi k}{N}\right)} = \frac{N^2}{2}, \quad \sum_{k=1}^{N-1} \frac{1}{1 - \cos\left(\frac{\pi k}{N}\right)} = \frac{2N^2 + 1}{6}, \quad \sum_{k=1}^{N-1} \frac{(-1)^k}{1 - \cos\left(\frac{\pi k}{N}\right)} = \frac{1 - N^2}{6}.$$

For the special case $|n-m| = N/2$ with N even Eq. (27) becomes

$$\widetilde{W}_n(n, z) = \frac{T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - 1}{zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} = \frac{2 \left[U_{\frac{N}{2}-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^2}{zq U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} \quad (28)$$

PROPAGATOR WITH ONE SHORTCUT AND ONE ABSORBING SITE

AUDIT NOTE: model statement required before the defect formula

The following original paragraph begins with a probability-conserving shortcut defect formula. That formula is valid when the shortcut transfers probability from one transient site to another transient site. In the present use case, however, the shortcut endpoint is also the absorbing site. Therefore one must state the transient-state model before applying any rank-one identity.

Let $\lambda = \beta(1 - q)$ and let P_0 be the transition matrix on the nonabsorbing transient sites for the ring with absorbing target v already removed. A shortcut $u \rightarrow v$ replaces a fraction λ of the stay-put mass at u by immediate absorption. Hence, in transient space,

$$P = P_0 - \lambda|u\rangle\langle u|. \quad (29)$$

This sign is the key point. The perturbation is negative because probability leaves the transient set at u .

We start from the general propagator with a shortcut from $n = u$ to site $n = v$ and we have

$$\tilde{S}_{n_0}(n, z) = \tilde{P}_{n_0}(n, z) + \frac{z\beta(1 - q)\tilde{P}_{n_0}(u, z) [\tilde{P}_u(n, z) - \tilde{P}_v(n, z)]}{1 - z\beta(1 - q) [\tilde{P}_u(u, z) - \tilde{P}_v(u, z)]}, \quad (30)$$

where $\tilde{P}_r(n, z)$ is the ring propagator generating function without shortcuts. If the ring also has an absorbing site, the modularity of the defect technique allows to replace the (probability conserving) propagator $\tilde{P}_r(n, z)$ with the propagator $\tilde{W}_r(n, z)$ in Eq. (17)

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) + \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z) [\tilde{W}_u(n, z) - \tilde{W}_v(n, z)]}{1 - z\beta(1 - q) [\tilde{W}_u(u, z) - \tilde{W}_v(u, z)]}. \quad (31)$$

When the absorbing site coincides with the shortcut site, i.e. $m = v$, $\tilde{W}_v(n, z) = 0$ and Eq. (31) simplifies further to

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) + \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 - z\beta(1 - q)\tilde{W}_u(u, z)}. \quad (32)$$

ERROR and CORRECTION: shortcut to the absorbing target is killing

ERROR. Eq. (32) is not valid for the stochastic interpretation in which the shortcut endpoint v is absorbing. The original formula has a positive rank-one correction and the denominator $1 - z\beta(1 - q)\tilde{W}_u(u, z)$. This would increase transient occupation, which is the opposite of a shortcut that sends probability to an absorbing target.

REDLINE. The core local edit in Eq. (32) is

$$+ \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 - z\beta(1 - q)\tilde{W}_u(u, z)} \implies - \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}. \quad (33)$$

CORRECTION. Starting from Eq. (29),

$$\begin{aligned} (I - zP)^{-1} &= (I - zP_0 + z\lambda|u\rangle\langle u|)^{-1} \\ &= (I - zP_0)^{-1} - \frac{z\lambda(I - zP_0)^{-1}|u\rangle\langle u|(I - zP_0)^{-1}}{1 + z\lambda\langle u|(I - zP_0)^{-1}|u\rangle}. \end{aligned} \quad (34)$$

Therefore the corrected transient propagator is

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}. \quad (35)$$

WHY. The plus sign in the denominator and the minus sign in the correction are forced by the Sherman–Morrison identity applied to $P = P_0 - \lambda|u\rangle\langle u|$.

After summing Eq. (31) over n to obtain the survival probability $\tilde{S}_{n_0}(z) = [1 - \tilde{F}_{n_0}(v, z)] / (1 - z)$, one expresses the generating function of the first-passage probability to site v as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) - \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 - z\beta(1-q)\tilde{W}_u(u, z)}, \quad (36)$$

or more explicitly as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) - qz\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)] \tilde{H}_{N,|u-v|}(z), \quad (37)$$

ERROR and CORRECTION: first-passage sign

ERROR. Eqs. (36)–(37) inherit the wrong sign from Eq. (32). They make the shortcut contribution negative. For $0 \leq z < 1$, the factors $\tilde{W}_{n_0}(u, z)$ and $1 - \tilde{\mathcal{F}}_u(v, z)$ have non-negative coefficients, so a shortcut into the target should increase the hitting generating function.

REDLINE. The original first-passage correction should be crossed out and replaced as

$$\begin{aligned} \tilde{\mathcal{F}}_{n_0}(v, z) - \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 - z\beta(1-q)\tilde{W}_u(u, z)} \\ \Rightarrow \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \end{aligned} \quad (38)$$

CORRECTION. Summing Eq. (35) over all transient sites gives

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (39)$$

Equivalently,

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)] \tilde{H}_{N,|u-v|}^{\text{corr}}(z), \quad (40)$$

where the corrected auxiliary factor is defined below.

with $\tilde{W}_{n_0}(u, z)$ and $\tilde{\mathcal{F}}_r(m, z)$ defined, respectively, in Eqs. (19) and (13), and

$$\tilde{H}_{N,|u-v|}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 - z\beta(1-q)\tilde{W}_u(u, z)}. \quad (41)$$

CORRECTION: auxiliary denominator

CORRECTION. The corrected auxiliary factor corresponding to Eq. (40) is

$$\frac{1}{1 - z\beta(1-q)\tilde{W}_u(u, z)} \Rightarrow \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (42)$$

Thus

$$\tilde{H}_{N,|u-v|}^{\text{corr}}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (43)$$

This is the denominator change that propagates through the rest of the shortcut analysis.

We now consider the symmetric case $|u - v| = N/2$ with N even, and use $\widetilde{\mathcal{W}}_u(u, z)$ from Eq. (28) in $\widetilde{W}_u(u, z)$, that allows to simplify Eq. (41) to

$$\widetilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} - \phi(z)}, \quad (44)$$

ERROR and CORRECTION: symmetric-case pole geometry

ERROR. Eq. (44) uses $q/[\beta(1-q)] - \phi(z)$ because it came from the wrong-sign transient resolvent. Consequently the later figure analysis based on a positive horizontal line intersecting $\phi(1/s)$ is not the pole geometry of the corrected stochastic model.

REDLINE. The denominator-level replacement is

$$\frac{1}{\frac{q}{\beta(1-q)} - \phi(z)} \Rightarrow \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}. \quad (45)$$

CORRECTION. In the symmetric case the auxiliary factor is

$$\widetilde{H}_{N,N/2}^{\text{corr}}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}. \quad (46)$$

Equivalently, if the same plotted curve $\phi(1/s)$ is retained, the horizontal line is $-q/[\beta(1-q)]$, not $+q/[\beta(1-q)]$.

CORRECTED CONTINUATION. Let $N = 2L$, $v = 0$, $u = L$, and let ρ be the starting distance from v along either half-ring. Define

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right). \quad (47)$$

The corrected first-passage generating function is

$$\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad U_{-1} = 0. \quad (48)$$

This formula replaces the downstream pole analysis based on Eq. (44).

where

$$\phi(z) = \frac{2U_{N/2-1}^2 \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (49)$$

whose time-dependence is $H_{N,N/2}(0) = \beta(1-q)/q$ and for $t \geq 1$ is given by the complex integral (after the change of integration variable $z = 1/s$ and the region of integration becoming $|s| \leq 1$)

$$H_{N,N/2}(t) = \frac{1}{2\pi i} \oint_{|s| \leq 1} \frac{s^{t-1}}{\frac{q}{\beta(1-q)} - \phi\left(\frac{1}{s}\right)}. \quad (50)$$

The denominator in the integrand of Eq. (50) has M real zeros s_k ($k = 1, 2, \dots, M$) and are shown graphically in Fig. 1 (see the figure caption for an explanation of when $M = N/2$ and $M = N/2 - 1$).

AUDIT NOTE: original Fig. 1 is downstream of the wrong sign

The original caption is preserved above. Under the corrected model, the intersection picture must be redrawn with the horizontal line $+q/[\beta(1-q)]$ replaced by $-q/[\beta(1-q)]$. Also note the typographical error in the original caption: it says “~~given the requirement~~ $q > 1$ ”, but the stochastic model requires $0 < q < 1$ and a nonzero shortcut probability proportional to $1 - q$ requires $q < 1$.

Original figure file `phifunction.pdf` was not included with the Outlook TeX attachment. The annotated copy keeps the original figure call as a portable placeholder.

FIG. 1. Graphical representation of the poles of Eq. (44) for the case $N = 10$, $q = 2/3$ and $\beta = 2/7$ which are the values of s in the range $-1 \leq s \leq 1$ where the horizontal dotted line $\frac{q}{\beta(1-q)}$ intercepts the function $\phi(s^{-1})$. Depending on N , β and q (N even, $0 < \beta \leq 1$ and $0 < q < 1$) one may have $M = \frac{N}{2} - 1$ poles, when $\frac{q}{\beta(1-q)} < \frac{N}{2}$, or $M = \frac{N}{2}$ poles, when $\frac{q}{\beta(1-q)} \geq \frac{N}{2}$, which is the case displayed above. The asymptotes of the $\phi(s^{-1})$ function are given from right to left by, respectively, α_k in Eq. (12) with $k = 1, 2, \dots, \frac{N}{2}$, and they are interlaced with the poles s_k with $k = 1, \dots, M$. We follow the same principle and order from right to left the poles s_k with increasing k , thus with s_1 representing the largest positive pole. With $q = 1$ the curves are symmetric around the origin, but given the requirement $q > 1$ to have a non-zero probability to move from site u to site v , the function $\phi(s^{-1})$ is never symmetric around $s = 0$, which implies that in magnitude one has $|s_{\frac{N}{2}}| < s_1$ for any $q < 1$. The paired tick marks without numerical values on the top and bottom horizontal axes correspond, from right to left, to the values γ_k in Eq. (22) with $k = 1, 2, \dots, \frac{N}{2}$, which are interlaced with the α_k (the values for $N/2 + 1 \leq k \leq N - 1$ repeat themselves).

s_k (real) poles in the integrand with $k = 1, \dots, N/2 - 1$ in the dominator ($-1 < s_k < 1$) can be found numerically as

$$H_{N,N/2}(t) = \sum_{k=1}^M c_k s_k^{t-1} \quad (51)$$

with

$$c_k = \frac{2q}{N} \frac{\left\{ \left(1 + \frac{1}{q} [s_k - 1]\right)^2 - 1 \right\} \left\{ T_{\frac{N}{2}} \left(1 + \frac{1}{q} [s_k - 1]\right) \right\}^2}{\left(1 + \frac{1}{N}\right) \left(1 + \frac{1}{q} [s_k - 1]\right) U_{N-1} \left(1 + \frac{1}{q} [s_k - 1]\right) - T_N \left(1 + \frac{1}{q} [s_k - 1]\right) - U_{N-2} \left(1 + \frac{1}{q} [s_k - 1]\right) - 1}. \quad (52)$$

Analysis of c_k shows that $c_k \geq 0$ for any real s_k and one may have $c_k = 0$ only when $s_k = \alpha_k$. As α_k are the asymptotes of $\phi(1/s)$, then we conclude that $c_k > 0$ for all k .

The time dependence of Eq. (37) for $|u - v| = N/2$ is thus given by the time-convoluted expression for $t \geq 1$

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) - q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t',t''} - \mathcal{F}_u(v, t' - t'')] H_{N,N/2}(t'') \quad (53)$$

and $F_{n_0}(v, 0) = 0$. Using Eqs. (10), (21) and (51), after performing the various time convolutions, one obtains

$$\begin{aligned} F_{n_0}(v, t) = & \sum_{\ell=1}^{\frac{N}{2}} f_{\ell}(n_0, u) \alpha_{\ell}^{t-1} - q \sum_{k=1}^M \left\{ \sum_{r=1}^{N-1} \frac{c_k G_r^{(1)}(n_0, u, v)}{s_k(s_k - \gamma_r)} (s_k^t - \gamma_r^t) + \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k g_{\ell}^{(2)}(n_0, u, v)}{s_k(s_k - \alpha_{\ell})} (s_k^t - \alpha_{\ell}^t) \right\} \\ & + q \sum_{k=1}^M \sum_{r=1}^{N-1} \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k G_r^{(1)}(n_0, u, v) f_{\ell}(u, v)}{s_k - \alpha_{\ell}} \left[\frac{s_k^t - \gamma_r^t}{\alpha_{\ell}(s_k - \gamma_r)} - \frac{\alpha_{\ell}^t - \gamma_r^t}{s_k(\alpha_{\ell} - \gamma_r)} \right] - q \sum_{k=1}^M \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k g_{\ell}^{(2)}(n_0, u, v) f_{\ell}(u, v)}{s_k \alpha_{\ell}(s_k - \alpha_{\ell})} \alpha_{\ell}^t t \\ & + q \underbrace{\sum_{k=1}^M \sum_{r=1}^{\frac{N}{2}} \sum_{\ell=1}^{\frac{N}{2}}}_{\ell \neq r} \frac{c_k g_r^{(2)}(n_0, u, v) f_{\ell}(u, v)}{s_k \alpha_{\ell}(s_k - \alpha_{\ell})} \left[s_k \frac{s_k^t - \gamma_r^t}{s_k - \gamma_r} - \alpha_{\ell} \frac{\alpha_{\ell}^t - \gamma_r^t}{\alpha_{\ell} - \gamma_r} \right]. \end{aligned} \quad (54)$$

ERROR and CORRECTION: downstream convolution should be replaced

ERROR. Eq. (54) is downstream of the wrong $\tilde{H}$ denominator, so it should not be used as the final time-domain answer. There are also local bookkeeping problems: the baseline no-shortcut first-passage term should point to v , not u , and the odd-sector propagator terms involving $g_r^{(2)}$ should be paired with the odd-sector eigenvalue α_r rather than γ_r .

REDLINE. Even before replacing the whole expression, two local edits that a reader should notice are

$$\cancel{f_{\ell}(n_0, u)} \implies f_{\ell}(n_0, v), \quad \cancel{\gamma_r \text{ on an odd-sector } g_r^{(2)} \text{ term}} \implies \alpha_r. \quad (55)$$

CORRECTION. Use the corrected closed form Eq. (48) and expand it by residues. Let

$$D(y) = aT_L(y) + U_{L-1}(y), \quad N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y). \quad (56)$$

If y_j are the roots of D and $s_j = 1 - q + qy_j$, then

$$F_\rho(t) = \sum_{j=1}^L B_{\rho j} s_j^{t-1}, \quad B_{\rho j} = \frac{q N_\rho(y_j)}{D'(y_j)}. \quad (57)$$

This is shorter and keeps the coefficients tied directly to the corrected pole equation.

The exponentially decaying dominant term in the first-passage probability can now be evinced by finding the largest positive coefficient among α_k , γ_k and s_k with $k = 1$ identifying the largest value for each one. As $\alpha_1 > \gamma_1$, it suffices to compare α_1 and s_1 . As graphically in Fig. 1 we have observed that when $q/[\beta(1-q)] \geq N/2$, the largest coefficient is the pole s_1 , whereas it is α_1 if $q/[\beta(1-q)] < N/2$, we are able to identify analytically the necessary condition for the second peak to be present, and that is $\beta \leq \frac{2q}{(1-q)N}$. From the last term in the second line of Eq. (54) one may also notice that when $\beta > \frac{2q}{(1-q)N}$ the long time dependence is dominated by $\alpha_1^t t$, which indicates a logarithmic correction that slows the exponential decay. In other words the introduction of the shortcut between u and v changes the long time dependence of the first-passage probability from α_1^t to either $\alpha_1^t t$ when $\beta \geq \frac{2q}{(1-q)N}$ or to s_1^t when $\beta < \frac{2q}{(1-q)N}$. While the case $\beta = \frac{2q}{(1-q)N}$ corresponds to the transition from $\alpha_1^t t$ to s_1^t , $s_1 = 1$ and thus it does not affect the long time dependence.

ERROR and CORRECTED CONCLUSION: tail asymptotics vs finite-time bimodality

ERROR. The preceding original paragraph is the main conclusion that changes. The claimed threshold

$$\beta \leq \frac{2q}{(1-q)N}$$

comes from the wrong-sign denominator in Eq. (44). It is not a valid analytical threshold for the corrected stochastic shortcut-to-absorbing target model. The asserted $t\alpha_1^t$ term is a repeated-pole artefact of the uncorrected pole collision.

REDLINE. The conclusion-level replacement is

$$\beta \leq \frac{2q}{(1-q)N} \text{ as corrected condition} \implies \text{no such tail-dominance threshold follows from the corrected model}, \quad (58)$$

and

$$t\alpha_1^t \text{ or } s_1^t \implies B_{\rho 1} s_1^{t-1} \text{ for the corrected right tail.} \quad (59)$$

CORRECTED CONTINUATION. The corrected pole equation is

$$aT_L(y_s) + U_{L-1}(y_s) = 0, \quad y_s = 1 + \frac{s-1}{q}. \quad (60)$$

Equivalently, writing $y_s = \cos \theta$,

$$\tan(L\theta) = -a \sin \theta. \quad (61)$$

The largest positive root satisfies

$$\gamma_1 < s_1 < \alpha_1, \quad \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}, \quad \alpha_1 = 1 - q + q \cos \frac{\pi}{N}. \quad (62)$$

Therefore the corrected right tail is

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty, \quad (63)$$

not generically $t\alpha_1^t$.

ADDED DECAY DEPENDENCE. For weak shortcut strength, perturbing the corrected pole equation gives

$$s_1 = \alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2), \quad (64)$$

and hence

$$1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2). \quad (65)$$

For large N this becomes

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}. \quad (66)$$

FINITE-TIME BIMODALITY. Tail dominance is not the same question as whether the finite-time first-passage curve has two visually resolved peaks. The corrected formula above determines the asymptotic right tail. A finite-time double peak should be checked from the corrected finite-time series, with an explicit peak/valley criterion.

CORRECTED REPLACEMENT PARAGRAPH FOR THE AUTHOR

The shortcut-to-target calculation should be rewritten as follows. In the transient state space, a shortcut from u to the absorbing site v is an additional killing channel at u , so the transient matrix is $P = P_0 - \beta(1-q)|u\rangle\langle u|$. The rank-one resolvent identity then gives the denominator $1 + z\beta(1-q)\widetilde{W}_u(u, z)$, not $1 - z\beta(1-q)\widetilde{W}_u(u, z)$. Consequently the corrected symmetric first-passage generating function is

$$\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right).$$

The corrected poles solve $aT_L(y_s) + U_{L-1}(y_s) = 0$, or $\tan(L\theta) = -a \sin \theta$ when $y_s = \cos \theta$. The largest positive pole satisfies $\gamma_1 < s_1 < \alpha_1$, so the right tail is $F_\rho(t) \sim B_{\rho 1} s_1^{t-1}$ rather than a generic $t\alpha_1^t$ term. The original condition $\beta \leq 2q/[(1-q)N]$ is therefore not a corrected analytical condition for bimodality; finite-time bimodality must be assessed from the corrected finite-time series using an explicit peak/valley diagnostic.

Audited Corrected Analytical Derivation

Nearest-neighbour ring walk with one absorbing site and one shortcut

June 1, 2026

Contents

1	Purpose and final conclusions	2
2	Model and notation	3
3	Finite-sum identities with details	4
3.1	Proof of the central identity	4
3.2	The $y \rightarrow 1$ identity	5
3.3	The $y = \cos(2\pi r/N)$ identity	6
3.4	The reciprocal Fourier identity	6
4	Absorbing ring without shortcut	7
4.1	Probability-conserving ring propagator	7
4.2	First-passage probability to a single absorbing site	7
4.3	Survival probability	8
4.4	Renewal equation and the killed propagator	8
4.5	Boundary and initial-condition checks	10
4.6	Generating-function form	10
4.7	Special symmetric return propagator	11
5	Shortcut into the absorbing site	12
5.1	Transient matrix derivation of the sign	12
5.2	First-passage generating function	12
5.3	Why the sign is also forced physically	13
6	Symmetric shortcut: exact corrected first-passage generating function	13
6.1	Difference equation	13
6.2	Solving the recurrence	14
6.3	Checks on the symmetric formula	15
7	Poles, tail decay, and the corrected dependence on β	15
7.1	Pole equation	15
7.2	Location of the largest pole	16
7.3	Partial fractions and the exact tail coefficient	16
7.4	Weak-shortcut expansion	17

7.5 Strong-shortcut endpoint expansion	18
8 Consequences for the bimodality discussion	18
9 Audit notes: issues found in the original calculation	19
9.1 Substantive mathematical issues	19
9.2 Typographical and presentation issues	20
10 Optional numerical check for the figure parameters	21

1 Purpose and final conclusions

This document is a detailed audit and correction of the analytical calculation for a nearest-neighbour discrete-time random walk on a ring with one absorbing site and one directed shortcut into the absorbing site. The main conclusion is that the absorbing-ring part of the calculation can be made correct after several index, sign, and orthogonality corrections, but the shortcut-to-target part must be changed more substantially.

The key point is the following. If a shortcut from u to the absorbing target v replaces a fraction β of the stay-put probability $1 - q$, then in the transient state space the shortcut is an extra killing channel at u . Writing

$$\lambda = \beta(1 - q), \quad 0 \leq \beta \leq 1, \quad (1)$$

the corrected resolvent is

$$\boxed{\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\lambda \tilde{W}_{n_0}(u, z) \tilde{W}_u(n, z)}{1 + z\lambda \tilde{W}_u(u, z)}}. \quad (2)$$

The minus sign and the plus sign in the denominator are both forced by the stochastic interpretation: the shortcut moves probability mass from the transient site u to the absorbing site v .

Consequently, the first-passage generating function is

$$\boxed{\tilde{F}_{n_0}(v, z) = \tilde{F}_{n_0}^{(0)}(v, z) + \frac{z\lambda \tilde{W}_{n_0}(u, z) [1 - \tilde{F}_u^{(0)}(v, z)]}{1 + z\lambda \tilde{W}_u(u, z)}}. \quad (3)$$

Here the superscript (0) means “without the shortcut but with the absorbing target present.” This formula also gives a useful monotonicity check: the shortcut contribution to $\tilde{F}_{n_0}(v, z)$ has non-negative coefficients for $z \in [0, 1)$, so adding a shortcut to the target cannot decrease the probability of having hit the target by any fixed time.

In the symmetric case

$$N = 2L, \quad v = 0, \quad u = L, \quad (4)$$

and for an initial site at distance ρ from v along either half-ring, $0 \leq \rho \leq L$, the corrected first-passage generating function reduces to the closed form

$$\boxed{\tilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right)}. \quad (5)$$

We use the convention $U_{-1} = 0$. Formula (5) is the cleanest corrected analytical answer.

The poles are the roots of

$$\boxed{aT_L(y_s) + U_{L-1}(y_s) = 0, \quad y_s = 1 + \frac{s-1}{q}.} \quad (6)$$

Equivalently, if $y_s = \cos \theta$, the roots satisfy

$$\boxed{\tan(L\theta) = -a \sin \theta.} \quad (7)$$

The largest positive root s_1 lies in

$$\boxed{\gamma_1 < s_1 < \alpha_1,} \quad \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}, \quad \alpha_1 = 1 - q + q \cos \frac{\pi}{N}. \quad (8)$$

Thus the shortcut-to-target makes the long-time tail decay faster than the no-shortcut absorbing-ring tail. The original condition

$$\beta \leq \frac{2q}{(1-q)N} \quad (9)$$

is not a valid corrected condition for bimodality or pole dominance. The corrected long-time decay is

$$\boxed{F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty,} \quad (10)$$

not $t\alpha_1^t$. For weak shortcuts,

$$\boxed{s_1 = \alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2),} \quad (11)$$

and therefore

$$\boxed{1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2).} \quad (12)$$

For large N this becomes

$$\boxed{1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}.} \quad (13)$$

Equation (13) is the most direct corrected decay-dependence statement for the right tail of the second peak, if a second peak is present.

2 Model and notation

The ring sites are $0, 1, \dots, N-1$, with all site labels understood modulo N . The unmodified nearest-neighbour transition rule is

$$X_{t+1} = \begin{cases} X_t, & \text{with probability } 1-q, \\ X_t + 1, & \text{with probability } q/2, \\ X_t - 1, & \text{with probability } q/2, \end{cases} \quad 0 < q < 1. \quad (14)$$

The absorbing target is denoted by m or, in the shortcut section, by v . If m is absorbing, then once the walker reaches m it is removed from the transient dynamics. The killed propagator is

$$W_{n_0}(n, t) = \Pr\{X_t = n, X_0 = n_0, X_s \neq m \text{ for all } 0 \leq s \leq t\}, \quad (15)$$

for transient sites $n, n_0 \neq m$. The first-passage probability to m is

$$\mathcal{F}_{n_0}(m, t) = \Pr\{X_t = m, X_s \neq m \text{ for } 0 \leq s < t\}, \quad t \geq 1, \quad (16)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$ when $n_0 \neq m$.

Generating functions are defined by

$$\widetilde{W}_{n_0}(n, z) = \sum_{t=0}^{\infty} W_{n_0}(n, t) z^t, \quad \widetilde{F}_{n_0}(m, z) = \sum_{t=0}^{\infty} \mathcal{F}_{n_0}(m, t) z^t. \quad (17)$$

The frequently used change of variable is

$$y = y(z) = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right). \quad (18)$$

It is useful because

$$1 - z[1 - q + q \cos \theta] = zq[y - \cos \theta]. \quad (19)$$

When $N = 2L$ is even, the odd-sector eigenvalues are

$$\alpha_\ell = 1 - q + q \cos \frac{(2\ell - 1)\pi}{N}, \quad \ell = 1, \dots, L, \quad (20)$$

and the even-sector ring eigenvalues are

$$\gamma_r = 1 - q + q \cos \frac{2\pi r}{N}, \quad r = 1, \dots, N - 1. \quad (21)$$

3 Finite-sum identities with details

Let

$$\theta_j = \frac{\pi j}{N}, \quad c_j = \cos \theta_j, \quad A_j(d) = [1 - (-1)^j] \sin(d\theta_j) \sin \theta_j, \quad (22)$$

where $0 < d < N$ is an integer. The central identity is

$$\frac{1}{N} \sum_{j=1}^{N-1} \frac{A_j(d)}{y - c_j} = \frac{U_{N-d-1}(y) + U_{d-1}(y)}{U_{N-1}(y)}. \quad (23)$$

Here T_n and U_n are Chebyshev polynomials of the first and second kind, with

$$T_n(\cos \theta) = \cos(n\theta), \quad U_n(\cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}, \quad U_{-1}(y) = 0. \quad (24)$$

3.1 Proof of the central identity

The denominator $U_{N-1}(y)$ has the simple roots

$$y = c_j = \cos \frac{\pi j}{N}, \quad j = 1, \dots, N - 1. \quad (25)$$

Indeed,

$$U_{N-1}(c_j) = \frac{\sin(N\theta_j)}{\sin \theta_j} = \frac{\sin(\pi j)}{\sin \theta_j} = 0. \quad (26)$$

We compute the residue of the right-hand side of (23) at $y = c_j$. The numerator is

$$\begin{aligned} U_{N-d-1}(c_j) + U_{d-1}(c_j) &= \frac{\sin((N-d)\theta_j) + \sin(d\theta_j)}{\sin \theta_j} \\ &= \frac{\sin(N\theta_j - d\theta_j) + \sin(d\theta_j)}{\sin \theta_j}. \end{aligned} \quad (27)$$

Since $N\theta_j = \pi j$,

$$\begin{aligned} \sin(N\theta_j - d\theta_j) &= \sin(\pi j) \cos(d\theta_j) - \cos(\pi j) \sin(d\theta_j) \\ &= 0 - (-1)^j \sin(d\theta_j) = -(-1)^j \sin(d\theta_j). \end{aligned} \quad (28)$$

Substituting this into (27) gives

$$U_{N-d-1}(c_j) + U_{d-1}(c_j) = \frac{[1 - (-1)^j] \sin(d\theta_j)}{\sin \theta_j}. \quad (29)$$

Next, differentiate $U_{N-1}(\cos \theta) = \sin(N\theta)/\sin \theta$ with respect to y . Because $y = \cos \theta$, $dy/d\theta = -\sin \theta$. Thus

$$\frac{d}{dy} U_{N-1}(\cos \theta) = -\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\frac{\sin(N\theta)}{\sin \theta} \right]. \quad (30)$$

At $\theta = \theta_j$, $\sin(N\theta_j) = 0$ and $\cos(N\theta_j) = (-1)^j$, so

$$\begin{aligned} U'_{N-1}(c_j) &= -\frac{1}{\sin \theta_j} \left[\frac{N \cos(N\theta_j) \sin \theta_j - \sin(N\theta_j) \cos \theta_j}{\sin^2 \theta_j} \right] \\ &= -\frac{1}{\sin \theta_j} \left[\frac{N(-1)^j \sin \theta_j}{\sin^2 \theta_j} \right] = \frac{N(-1)^{j+1}}{\sin^2 \theta_j}. \end{aligned} \quad (31)$$

Therefore the residue of the right-hand side at $y = c_j$ is

$$\begin{aligned} \text{Res}_{y=c_j} \frac{U_{N-d-1}(y) + U_{d-1}(y)}{U_{N-1}(y)} &= \frac{U_{N-d-1}(c_j) + U_{d-1}(c_j)}{U'_{N-1}(c_j)} \\ &= \frac{[1 - (-1)^j] \sin(d\theta_j)}{\sin \theta_j} \frac{\sin^2 \theta_j}{N(-1)^{j+1}}. \end{aligned} \quad (32)$$

If j is even, the factor $1 - (-1)^j$ is zero. If j is odd, then $(-1)^{j+1} = 1$. Hence in all cases

$$\text{Res}_{y=c_j} \frac{U_{N-d-1}(y) + U_{d-1}(y)}{U_{N-1}(y)} = \frac{1}{N} [1 - (-1)^j] \sin(d\theta_j) \sin \theta_j = \frac{A_j(d)}{N}. \quad (33)$$

The left-hand side of (23) has exactly the same simple pole at $y = c_j$ and exactly the same residue $A_j(d)/N$. The difference between the two sides has no finite poles. Both sides tend to zero as $y \rightarrow \infty$. Therefore the difference is an entire rational function tending to zero at infinity, hence it is identically zero. This proves (23).

3.2 The $y \rightarrow 1$ identity

Set $y = 1$ in (23). Since $U_n(1) = n + 1$,

$$U_{N-d-1}(1) + U_{d-1}(1) = (N-d) + d = N, \quad U_{N-1}(1) = N. \quad (34)$$

Thus

$$\boxed{\frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d\theta_j) \sin \theta_j}{1 - \cos \theta_j} = 1.} \quad (35)$$

3.3 The $y = \cos(2\pi r/N)$ identity

For $r = 0, 1, \dots, N-1$, set

$$y = \cos \frac{2\pi r}{N}. \quad (36)$$

If this value is not a pole, the substitution is direct. If it is a zero of U_{N-1} , the value is the removable limit already determined by the residue calculation above. Using the trigonometric form gives

$$\boxed{\frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d\theta_j) \sin \theta_j}{\cos(2\pi r/N) - \cos \theta_j} = \cos \frac{2\pi r d}{N}}. \quad (37)$$

The case $r = 0$ is the same as (35) because $\cos(0) = 1$ and $\cos(0 \cdot d) = 1$.

3.4 The reciprocal Fourier identity

For $j = 1, \dots, N-1$ and $0 \leq d \leq N$, the second identity needed in the absorbing-propagator calculation is

$$\boxed{\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos(2\pi r d/N)}{\cos \theta_j - \cos(2\pi r/N)} = \frac{1}{N(1 - \cos \theta_j)} - \frac{\sin(d\theta_j)}{\sin \theta_j}}. \quad (38)$$

To see why this is the correct form, include temporarily the omitted $r = 0$ term. Since

$$\frac{1}{N} \frac{1}{\cos \theta_j - 1} = -\frac{1}{N(1 - \cos \theta_j)}, \quad (39)$$

(38) is equivalent to

$$\boxed{\frac{1}{N} \sum_{r=0}^{N-1} \frac{\cos(2\pi r d/N)}{\cos \theta_j - \cos(2\pi r/N)} = -\frac{\sin(d\theta_j)}{\sin \theta_j}}. \quad (40)$$

The right-hand side satisfies the homogeneous second-order recurrence

$$g_{d+1} - 2 \cos \theta_j g_d + g_{d-1} = 0, \quad g_d = -\frac{\sin(d\theta_j)}{\sin \theta_j}, \quad (41)$$

for $1 \leq d \leq N-1$. The Fourier sum on the left-hand side satisfies the same recurrence away from $d = 0$ because multiplication by $2 \cos(2\pi r/N)$ shifts the Fourier modes. The boundary condition is also consistent because $N\theta_j = \pi j$ implies

$$g_{N-d} = -\frac{\sin(N\theta_j - d\theta_j)}{\sin \theta_j} = -\frac{-(-1)^j \sin(d\theta_j)}{\sin \theta_j} = (-1)^j \frac{\sin(d\theta_j)}{\sin \theta_j}. \quad (42)$$

In the only use below, this identity is multiplied by $1 - (-1)^j$, so only odd j contributes, and for odd j the symmetry becomes $g_{N-d} = g_d$, as required by the cosine Fourier sum. This establishes (38) in the sector used by the calculation. Equivalently, one may derive it from the partial-fraction expansion of $[T_N(x) - 1]^{-1}$.

4 Absorbing ring without shortcut

4.1 Probability-conserving ring propagator

The probability-conserving ring propagator is

$$Q_{n_0}(n, t) = \frac{1}{N} \sum_{r=0}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} \gamma_r^t, \quad (43)$$

where $\gamma_0 = 1$ and γ_r for $r \geq 1$ is given by (21). Separating the zero mode gives

$$Q_{n_0}(n, t) = \frac{1}{N} + \frac{1}{N} \sum_{r=1}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} \gamma_r^t. \quad (44)$$

4.2 First-passage probability to a single absorbing site

Let the absorbing site be m and define

$$d_0 = |n_0 - m|, \quad 0 < d_0 < N. \quad (45)$$

The first-passage probability to m on the ring without shortcut is

$$\boxed{\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \lambda_j^{t-1}, \quad t \geq 1,} \quad (46)$$

where

$$\lambda_j = 1 - q + q \cos \theta_j. \quad (47)$$

If $N = 2L$, only the odd j values contribute, and (46) becomes

$$\mathcal{F}_{n_0}(m, t) = \sum_{\ell=1}^L f_\ell(n_0, m) \alpha_\ell^{t-1}, \quad (48)$$

with

$$\boxed{f_\ell(n_0, m) = \frac{2q}{N} \sin \frac{(2\ell - 1)\pi |n_0 - m|}{N} \sin \frac{(2\ell - 1)\pi}{N}.} \quad (49)$$

The generating function follows directly from (19):

$$\begin{aligned} \tilde{\mathcal{F}}_{n_0}(m, z) &= z \frac{q}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - z \lambda_j} \\ &= \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{y - \cos \theta_j} \\ &= \boxed{\frac{U_{N-d_0-1}(y) + U_{d_0-1}(y)}{U_{N-1}(y)}}. \end{aligned} \quad (50)$$

Setting $z = 1$ gives $y = 1$, and (35) gives

$$\tilde{\mathcal{F}}_{n_0}(m, 1) = 1. \quad (51)$$

Thus the absorbing target is eventually reached with probability one.

4.3 Survival probability

The survival probability up to and including time t is

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s). \quad (52)$$

Using (46), $\mathcal{F}_{n_0}(m, 0) = 0$, and the finite geometric sum

$$\sum_{s=1}^t \lambda_j^{s-1} = \frac{1 - \lambda_j^t}{1 - \lambda_j} = \frac{1 - \lambda_j^t}{q(1 - \cos \theta_j)}, \quad (53)$$

we get

$$\begin{aligned} \sum_{s=1}^t \mathcal{F}_{n_0}(m, s) &= \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} (1 - \lambda_j^t) \\ &= 1 - \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} \lambda_j^t, \end{aligned} \quad (54)$$

where the last step used (35). Hence

$$R_{n_0}(t) = \frac{1}{N} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} \lambda_j^t. \quad (55)$$

4.4 Renewal equation and the killed propagator

For $t \geq 1$, the killed propagator satisfies the renewal relation

$$W_{n_0}(n, t) = Q_{n_0}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0}(m, s) Q_m(n, t - s). \quad (56)$$

This subtracts all paths that first hit m at time s and then, in the unrestricted ring, continue from m to n in the remaining $t - s$ steps.

Insert (44) and (46). The zero Fourier mode of Q_m gives

$$\frac{1}{N} \sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} [1 - R_{n_0}(t)]. \quad (57)$$

This cancels the $1/N$ term in $Q_{n_0}(n, t)$ and leaves $R_{n_0}(t)/N$, which is precisely the second line in (59). For the nonzero Fourier modes, we need

$$\begin{aligned} \sum_{s=1}^t \lambda_j^{s-1} \gamma_r^{t-s} &= \gamma_r^{t-1} + \lambda_j \gamma_r^{t-2} + \cdots + \lambda_j^{t-1} \\ &= \frac{\gamma_r^t - \lambda_j^t}{\gamma_r - \lambda_j} \\ &= \frac{\gamma_r^t - \lambda_j^t}{q[\cos(2\pi r/N) - \cos \theta_j]}. \end{aligned} \quad (58)$$

Because this convolution is subtracted in (56), the denominator can be rewritten as

$$\cos \theta_j - \cos(2\pi r/N),$$

giving the fully expanded expression

$$\begin{aligned} W_{n_0}(n, t) = & \frac{1}{N} \sum_{r=1}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} \gamma_r^t \\ & + \frac{1}{N^2} \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} \lambda_j^t \\ & + \frac{1}{N^2} \sum_{j=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \cos \frac{2\pi r(n-m)}{N}}{\cos \theta_j - \cos(2\pi r/N)} (\gamma_r^t - \lambda_j^t). \end{aligned} \quad (59)$$

The sign of the last bracket is important. It must be $\gamma_r^t - \lambda_j^t$ when the denominator is $\cos \theta_j - \cos(2\pi r/N)$.

Now reduce the double sum. The part proportional to γ_r^t uses (37) with $d = d_0$:

$$\begin{aligned} & \sum_{j=1}^{N-1} \frac{[1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j}{\cos \theta_j - \cos(2\pi r/N)} \\ & = -N \cos \frac{2\pi r d_0}{N}. \end{aligned} \quad (60)$$

Therefore the γ_r^t part of the double sum is

$$-N \sum_{r=1}^{N-1} \cos \frac{2\pi r d_0}{N} \cos \frac{2\pi r(n-m)}{N} \gamma_r^t. \quad (61)$$

The part proportional to $-\lambda_j^t$ uses (38) with $d = |n - m|$:

$$\begin{aligned} & - \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \lambda_j^t \sum_{r=1}^{N-1} \frac{\cos(2\pi r(n-m)/N)}{\cos \theta_j - \cos(2\pi r/N)} \\ & = - \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d_0 \theta_j) \sin \theta_j \lambda_j^t \left[\frac{1}{1 - \cos \theta_j} - N \frac{\sin(d \theta_j)}{\sin \theta_j} \right] \\ & = - \sum_{j=1}^{N-1} [1 - (-1)^j] \left[\frac{\sin(d_0 \theta_j) \sin \theta_j}{1 - \cos \theta_j} - N \sin(d_0 \theta_j) \sin(d \theta_j) \right] \lambda_j^t. \end{aligned} \quad (62)$$

After substituting (61) and (62) into (59), the survival term cancels the first term inside the square brackets of (62). The final killed propagator is

$$\boxed{W_{n_0}(n, t) = \frac{1}{N} \sum_{r=1}^{N-1} \left[\cos \frac{2\pi r(n - n_0)}{N} - \cos \frac{2\pi r(n - m)}{N} \cos \frac{2\pi r(n_0 - m)}{N} \right] \gamma_r^t + \frac{1}{N} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin \frac{\pi j |n_0 - m|}{N} \sin \frac{\pi j |n - m|}{N} \lambda_j^t.} \quad (63)$$

4.5 Boundary and initial-condition checks

If $n = m$, then the sine factor in the second line of (63) is zero and the first-line bracket becomes

$$\cos \frac{2\pi r(m - n_0)}{N} - 1 \cdot \cos \frac{2\pi r(n_0 - m)}{N} = 0. \quad (64)$$

So $W_{n_0}(m, t) = 0$, as required for a killed propagator. Similarly, if $n_0 = m$, both lines vanish for transient n .

At $t = 0$, for transient $n, n_0 \neq m$, (63) gives the identity $W_{n_0}(n, 0) = \delta_{n, n_0}$. Here are the details. Let

$$d = |n - m|, \quad d_0 = |n_0 - m|. \quad (65)$$

The basic Fourier identity is

$$\frac{1}{N} \sum_{r=1}^{N-1} \cos \frac{2\pi r(n - n_0)}{N} = \delta_{n, n_0} - \frac{1}{N}. \quad (66)$$

The product-to-sum formula gives

$$\begin{aligned} \sum_{r=1}^{N-1} \cos \frac{2\pi r d}{N} \cos \frac{2\pi r d_0}{N} &= \frac{1}{2} \sum_{r=1}^{N-1} \left[\cos \frac{2\pi r(d - d_0)}{N} + \cos \frac{2\pi r(d + d_0)}{N} \right] \\ &= \frac{N}{2} (\delta_{d, d_0} + \delta_{d, N-d_0}) - 1, \end{aligned} \quad (67)$$

where the Kronecker deltas are interpreted modulo the ring. The odd sine sum is

$$\begin{aligned} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin(d\theta_j) \sin(d_0\theta_j) &= 2 \sum_{\substack{1 \leq j \leq N-1 \\ j \text{ odd}}} \sin(d\theta_j) \sin(d_0\theta_j) \\ &= \frac{N}{2} (\delta_{d, d_0} + \delta_{d, N-d_0}). \end{aligned} \quad (68)$$

Substituting (67) and (68) into (63) at $t = 0$ cancels the distance-only terms and leaves δ_{n, n_0} .

4.6 Generating-function form

Define

$$G_r^{(1)}(n_0, n, m) = \frac{1}{N} \left[\cos \frac{2\pi r(n - n_0)}{N} - \cos \frac{2\pi r(n - m)}{N} \cos \frac{2\pi r(n_0 - m)}{N} \right], \quad (69)$$

$$G_j^{(2)}(n_0, n, m) = \frac{1}{N} [1 - (-1)^j] \sin \frac{\pi j |n_0 - m|}{N} \sin \frac{\pi j |n - m|}{N}. \quad (70)$$

Then

$$W_{n_0}(n, t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0, n, m) \gamma_r^t + \sum_{j=1}^{N-1} G_j^{(2)}(n_0, n, m) \lambda_j^t. \quad (71)$$

If $N = 2L$, the second sum can be written over odd modes only:

$$W_{n_0}(n, t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0, n, m) \gamma_r^t + \sum_{\ell=1}^L g_\ell^{(2)}(n_0, n, m) \alpha_\ell^t, \quad (72)$$

where

$$g_\ell^{(2)}(n_0, n, m) = \frac{2}{N} \sin \frac{(2\ell-1)\pi |n_0 - m|}{N} \sin \frac{(2\ell-1)\pi |n - m|}{N}. \quad (73)$$

The generating function is therefore

$$\widetilde{W}_{n_0}(n, z) = \sum_{r=1}^{N-1} \frac{G_r^{(1)}(n_0, n, m)}{1 - z\gamma_r} + \sum_{\ell=1}^L \frac{g_\ell^{(2)}(n_0, n, m)}{1 - z\alpha_\ell}. \quad (74)$$

4.7 Special symmetric return propagator

Let $N = 2L$, $m = 0$, and $n = n_0 = L$. Then the first line of (63) vanishes because

$$\cos \frac{2\pi r(L-L)}{N} = 1, \quad \cos \frac{2\pi rL}{N} \cos \frac{2\pi rL}{N} = \cos^2(\pi r) = 1. \quad (75)$$

The second line becomes

$$W_L(L, t) = \frac{1}{N} \sum_{j=1}^{N-1} [1 - (-1)^j] \sin^2 \frac{\pi j}{2} \lambda_j^t. \quad (76)$$

Only odd $j = 2\ell - 1$ contribute. For odd j , $1 - (-1)^j = 2$ and $\sin^2(\pi j/2) = 1$, so

$$W_L(L, t) = \frac{2}{N} \sum_{\ell=1}^L \alpha_\ell^t. \quad (77)$$

Hence

$$\widetilde{W}_L(L, z) = \frac{2}{N} \sum_{\ell=1}^L \frac{1}{1 - z\alpha_\ell} = \frac{2}{N} \frac{1}{zq} \sum_{\ell=1}^L \frac{1}{y - \cos \frac{(2\ell-1)\pi}{2L}}. \quad (78)$$

The zeros of $T_L(y)$ are exactly $\cos[(2\ell-1)\pi/(2L)]$, so

$$\frac{T'_L(y)}{T_L(y)} = \sum_{\ell=1}^L \frac{1}{y - \cos \frac{(2\ell-1)\pi}{2L}}. \quad (79)$$

Since $T'_L(y) = LU_{L-1}(y)$ and $N = 2L$, (78) gives

$$\widetilde{W}_L(L, z) = \frac{2}{2L} \frac{1}{zq} \frac{LU_{L-1}(y)}{T_L(y)} = \frac{U_{L-1}(y)}{zq T_L(y)}. \quad (80)$$

Using $U_{2L-1}(y) = 2T_L(y)U_{L-1}(y)$, this is also

$$\boxed{\widetilde{W}_L(L, z) = \frac{2U_{L-1}^2(y)}{zq U_{2L-1}(y)}}. \quad (81)$$

At $z = 1$, $y = 1$, so

$$\widetilde{W}_L(L, 1) = \frac{U_{L-1}(1)}{qT_L(1)} = \frac{L}{q} = \frac{N}{2q}. \quad (82)$$

5 Shortcut into the absorbing site

5.1 Transient matrix derivation of the sign

Let K_0 be the transient transition matrix for the ring with absorbing target v , but without the shortcut. Its resolvent is

$$R_0(z) = (I - zK_0)^{-1}, \quad [R_0(z)]_{n,n_0} = \widetilde{W}_{n_0}(n, z). \quad (83)$$

A shortcut from u to the absorbing target v that replaces probability $\lambda = \beta(1 - q)$ of the stay-put move at u removes that probability from the transient state u . Therefore, in the transient subspace,

$$K = K_0 - \lambda|u\rangle\langle u|. \quad (84)$$

The resolvent with the shortcut is

$$R(z) = (I - zK)^{-1} = [I - zK_0 + z\lambda|u\rangle\langle u|]^{-1}. \quad (85)$$

Set

$$A = I - zK_0. \quad (86)$$

Then

$$R(z) = [A + z\lambda|u\rangle\langle u|]^{-1}. \quad (87)$$

Use the Sherman–Morrison identity

$$(A + |p\rangle\langle q|)^{-1} = A^{-1} - \frac{A^{-1}|p\rangle\langle q|A^{-1}}{1 + \langle q|A^{-1}|p\rangle}. \quad (88)$$

Here $|p\rangle = z\lambda|u\rangle$ and $\langle q| = \langle u|$. Thus

$$R(z) = R_0(z) - \frac{z\lambda R_0(z)|u\rangle\langle u|R_0(z)}{1 + z\lambda\langle u|R_0(z)|u\rangle}. \quad (89)$$

Taking the (n, n_0) matrix element gives the corrected propagator

$$\boxed{\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\lambda \widetilde{W}_{n_0}(u, z) \widetilde{W}_u(n, z)}{1 + z\lambda \widetilde{W}_u(u, z)}}. \quad (90)$$

This is the same as (2). The sign is not a convention: it is fixed by the fact that the target state v is absorbing and therefore absent from the transient subspace.

5.2 First-passage generating function

Sum (90) over all transient sites $n \neq v$. For the baseline killed walk,

$$\sum_{n \neq v} \widetilde{W}_r(n, z) = \frac{1 - \widetilde{F}_r^{(0)}(v, z)}{1 - z}. \quad (91)$$

Therefore

$$\sum_{n \neq v} \widetilde{S}_{n_0}(n, z) = \frac{1 - \widetilde{F}_{n_0}^{(0)}(v, z)}{1 - z} - \frac{z\lambda \widetilde{W}_{n_0}(u, z)}{1 + z\lambda \widetilde{W}_u(u, z)} \frac{1 - \widetilde{F}_u^{(0)}(v, z)}{1 - z}. \quad (92)$$

But the left-hand side is also $(1 - \widetilde{F}_{n_0}(v, z))/(1 - z)$. Multiplying by $1 - z$ and rearranging gives

$$\boxed{\widetilde{F}_{n_0}(v, z) = \widetilde{F}_{n_0}^{(0)}(v, z) + \frac{z\lambda \widetilde{W}_{n_0}(u, z) [1 - \widetilde{F}_u^{(0)}(v, z)]}{1 + z\lambda \widetilde{W}_u(u, z)}}. \quad (93)$$

5.3 Why the sign is also forced physically

For $0 \leq z < 1$, all coefficients of $\widetilde{W}_{n_0}(u, z)$ and $1 - \widetilde{F}_u^{(0)}(v, z)$ are non-negative, and $1 + z\lambda\widetilde{W}_u(u, z) > 0$. Hence the shortcut term in (93) is non-negative. Thus adding a shortcut into the absorbing site increases the first-passage generating function, as it should.

By contrast, a formula with

$$+ \frac{z\lambda\widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z)}{1 - z\lambda\widetilde{W}_u(u, z)} \quad (94)$$

in the propagator would increase transient survival and decrease the hitting generating function. It can also produce a positive real singularity when

$$z\lambda\widetilde{W}_u(u, z) = 1. \quad (95)$$

In the symmetric case, $\widetilde{W}_L(L, 1) = N/(2q)$, so the denominator at $z = 1$ would be

$$1 - \frac{\beta(1-q)N}{2q}, \quad (96)$$

which is exactly the source of the original threshold. Such a singularity is an artefact of the wrong sign; it is not present in the corrected stochastic shortcut-to-target model.

6 Symmetric shortcut: exact corrected first-passage generating function

We now set

$$N = 2L, \quad v = 0, \quad u = L. \quad (97)$$

Let $F_\rho(z)$ be the first-passage generating function to 0 from a site at distance ρ from 0 along either half-ring, where $0 \leq \rho \leq L$. By symmetry, $F_{L+k} = F_{L-k}$.

6.1 Difference equation

For $1 \leq \rho \leq L-1$, the first step gives

$$F_\rho(z) = z \left[(1-q)F_\rho(z) + \frac{q}{2}F_{\rho-1}(z) + \frac{q}{2}F_{\rho+1}(z) \right]. \quad (98)$$

Move all terms to one side and divide by $zq/2$:

$$F_{\rho+1}(z) - 2yF_\rho(z) + F_{\rho-1}(z) = 0, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right). \quad (99)$$

The absorbing boundary is

$$F_0(z) = 1. \quad (100)$$

At $\rho = L$, the shortcut replaces probability $\lambda = \beta(1-q)$ of the stay-put probability by an immediate jump to the target. Therefore

$$F_L(z) = z [\lambda + (1-q-\lambda)F_L(z) + qF_{L-1}(z)]. \quad (101)$$

Using $1 - q - \lambda = (1 - \beta)(1 - q)$ and writing

$$h = \frac{\lambda}{q} = \frac{\beta(1-q)}{q}, \quad a = \frac{1}{h} = \frac{q}{\beta(1-q)}, \quad (102)$$

the boundary equation becomes

$$\boxed{(y+h)F_L(z) - F_{L-1}(z) = h.} \quad (103)$$

6.2 Solving the recurrence

The recurrence (99) is the Chebyshev recurrence. The solution satisfying $F_0 = 1$ can be written as

$$F_\rho = T_\rho(y) + C U_{\rho-1}(y), \quad (104)$$

because $T_0 = 1$ and $U_{-1} = 0$. Hence

$$F_L = T_L + C U_{L-1}, \quad F_{L-1} = T_{L-1} + C U_{L-2}. \quad (105)$$

Substitute these into (103):

$$\begin{aligned} h &= (y + h)(T_L + C U_{L-1}) - (T_{L-1} + C U_{L-2}) \\ &= (y T_L - T_{L-1}) + C(y U_{L-1} - U_{L-2}) + h T_L + h C U_{L-1}. \end{aligned} \quad (106)$$

The needed Chebyshev identities are

$$y U_{L-1} - U_{L-2} = T_L, \quad y T_L - T_{L-1} = (y^2 - 1) U_{L-1}. \quad (107)$$

Using them in (106) gives

$$h = (y^2 - 1) U_{L-1} + C T_L + h T_L + h C U_{L-1}. \quad (108)$$

Therefore

$$C(T_L + h U_{L-1}) = h(1 - T_L) - (y^2 - 1) U_{L-1}, \quad (109)$$

and

$$C = \frac{h(1 - T_L) - (y^2 - 1) U_{L-1}}{T_L + h U_{L-1}}. \quad (110)$$

Substitution into (104) gives one correct form of the answer:

$$F_\rho = T_\rho + U_{\rho-1} \frac{h(1 - T_L) - (y^2 - 1) U_{L-1}}{T_L + h U_{L-1}}. \quad (111)$$

Multiplying numerator and denominator by $a = 1/h$ gives

$$F_\rho = \frac{a T_\rho T_L + T_\rho U_{L-1} + U_{\rho-1}(1 - T_L) - a(y^2 - 1) U_{\rho-1} U_{L-1}}{a T_L + U_{L-1}}. \quad (112)$$

Now use the two product identities

$$T_\rho T_L - (y^2 - 1) U_{\rho-1} U_{L-1} = T_{L-\rho}, \quad (113)$$

$$T_\rho U_{L-1} + U_{\rho-1}(1 - T_L) = U_{L-\rho-1} + U_{\rho-1}. \quad (114)$$

For example, if $y = \cos \theta$, then (113) reads

$$\cos(\rho\theta) \cos(L\theta) + \sin(\rho\theta) \sin(L\theta) = \cos((L - \rho)\theta), \quad (115)$$

and (114) follows from

$$\cos(\rho\theta) \frac{\sin(L\theta)}{\sin \theta} + \frac{\sin(\rho\theta)}{\sin \theta} [1 - \cos(L\theta)] = \frac{\sin((L - \rho)\theta) + \sin(\rho\theta)}{\sin \theta}. \quad (116)$$

Thus (112) becomes

$$\boxed{\tilde{F}_\rho(z) = \frac{a T_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{a T_L(y) + U_{L-1}(y)}}. \quad (117)$$

This is the same as (5).

6.3 Checks on the symmetric formula

First, $\rho = 0$ gives

$$\tilde{F}_0(z) = \frac{aT_L + U_{-1} + U_{L-1}}{aT_L + U_{L-1}} = 1, \quad (118)$$

as required for a walker already at the absorbing site. Second, in the no-shortcut limit $\beta \rightarrow 0$, we have $a \rightarrow \infty$, so

$$\tilde{F}_\rho(z) \rightarrow \frac{T_{L-\rho}(y)}{T_L(y)}. \quad (119)$$

This agrees with the absorbing-ring formula (50). Indeed, for $N = 2L$,

$$\frac{U_{2L-\rho-1} + U_{\rho-1}}{U_{2L-1}} = \frac{T_{L-\rho}}{T_L}, \quad (120)$$

which follows from the same trigonometric product identities.

Third, when $\rho = L$,

$$\tilde{F}_L(z) = \frac{a + U_{L-1}(y)}{aT_L(y) + U_{L-1}(y)}. \quad (121)$$

At $z = 1$, $y = 1$, so $T_L(1) = 1$ and $U_{L-1}(1) = L$, giving

$$\tilde{F}_L(1) = \frac{a + L}{a + L} = 1. \quad (122)$$

Thus absorption remains certain.

7 Poles, tail decay, and the corrected dependence on β

7.1 Pole equation

Write $s = 1/z$, so

$$y_s = 1 + \frac{s - 1}{q}. \quad (123)$$

The poles of (117) are roots of

$$D(y_s) = aT_L(y_s) + U_{L-1}(y_s) = 0. \quad (124)$$

If $y_s = \cos \theta$, then

$$T_L(y_s) = \cos(L\theta), \quad U_{L-1}(y_s) = \frac{\sin(L\theta)}{\sin \theta}. \quad (125)$$

Therefore (124) becomes

$$a \cos(L\theta) + \frac{\sin(L\theta)}{\sin \theta} = 0. \quad (126)$$

Multiplying by $\sin \theta$ and dividing by $\cos(L\theta)$ away from the asymptotes gives

$$\boxed{\tan(L\theta) = -a \sin \theta.} \quad (127)$$

The corresponding pole value is

$$\boxed{s = 1 - q + q \cos \theta.} \quad (128)$$

7.2 Location of the largest pole

The no-shortcut largest odd-sector eigenvalue is

$$\alpha_1 = 1 - q + q \cos \frac{\pi}{2L} = 1 - q + q \cos \frac{\pi}{N}. \quad (129)$$

The largest even-sector nontrivial eigenvalue is

$$\gamma_1 = 1 - q + q \cos \frac{\pi}{L} = 1 - q + q \cos \frac{2\pi}{N}. \quad (130)$$

Consider the interval

$$\frac{\pi}{2L} < \theta < \frac{\pi}{L}. \quad (131)$$

In this interval,

$$\cos(L\theta) < 0, \quad \sin(L\theta) > 0, \quad \sin \theta > 0. \quad (132)$$

Hence $D(\cos \theta) = a \cos(L\theta) + \sin(L\theta)/\sin \theta$ tends to $U_{L-1} > 0$ at the left endpoint $\theta = \pi/(2L)$, while it equals $a \cos \pi = -a < 0$ at the right endpoint $\theta = \pi/L$. By continuity there is a root θ_1 in this interval. This root gives

$$\cos \frac{\pi}{L} < \cos \theta_1 < \cos \frac{\pi}{2L}, \quad (133)$$

and therefore

$$\boxed{\gamma_1 < s_1 < \alpha_1}. \quad (134)$$

Since the transient transition matrix is non-negative and primitive for $0 < q < 1$, its spectral radius is the unique positive dominant eigenvalue. Thus s_1 controls the long-time tail.

7.3 Partial fractions and the exact tail coefficient

Let

$$N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y), \quad D(y) = aT_L(y) + U_{L-1}(y). \quad (135)$$

Let y_k be a simple root of $D(y)$ and define

$$s_k = 1 - q + qy_k. \quad (136)$$

Near $s = s_k$,

$$\tilde{F}_\rho(1/s) = \frac{N_\rho(y_s)}{D(y_s)} = \frac{N_\rho(y_k)}{D'(y_k)(y_s - y_k)} + \text{regular terms}. \quad (137)$$

Since $y_s - y_k = (s - s_k)/q$, the residue in the s -plane is

$$\text{Res}_{s=s_k} \tilde{F}_\rho(1/s) = \frac{qN_\rho(y_k)}{D'(y_k)}. \quad (138)$$

The partial-fraction term corresponding to a pole s_k is

$$\frac{B_{\rho k}}{1 - zs_k}, \quad (139)$$

whose $s = 1/z$ form is

$$\frac{B_{\rho k}}{1 - s_k/s} = \frac{B_{\rho k}s}{s - s_k}. \quad (140)$$

Its residue at $s = s_k$ is $B_{\rho k} s_k$. Hence the coefficient multiplying s_k^t is

$$B_{\rho k}^{(z^t)} = \frac{q N_\rho(y_k)}{s_k D'(y_k)}. \quad (141)$$

Equivalently, if we write the time dependence as s_k^{t-1} for $t \geq 1$, then

$$F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \quad B_{\rho k} = \frac{q N_\rho(y_k)}{D'(y_k)}. \quad (142)$$

In particular,

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad B_{\rho 1} = \frac{q N_\rho(y_1)}{D'(y_1)}. \quad (143)$$

The decay time can be reported as

$$\tau_\beta = -\frac{1}{\log s_1}. \quad (144)$$

7.4 Weak-shortcut expansion

For small β , equivalently large

$$a = \frac{q}{\beta(1-q)}, \quad (145)$$

the largest root lies close to

$$\theta_0 = \frac{\pi}{2L}. \quad (146)$$

Write

$$\theta_1 = \theta_0 + \delta, \quad 0 < \delta \ll 1. \quad (147)$$

Then

$$L\theta_1 = \frac{\pi}{2} + L\delta, \quad \tan(L\theta_1) = \tan\left(\frac{\pi}{2} + L\delta\right) = -\cot(L\delta). \quad (148)$$

For small δ ,

$$-\cot(L\delta) = -\frac{1}{L\delta} + O(L\delta). \quad (149)$$

The root equation (127) gives

$$-\frac{1}{L\delta} + O(\delta) = -a \sin(\theta_0 + \delta). \quad (150)$$

To leading order,

$$\delta = \frac{1}{aL \sin \theta_0} + O(a^{-2}). \quad (151)$$

Now expand $s_1 = 1 - q + q \cos(\theta_0 + \delta)$:

$$\begin{aligned} s_1 &= 1 - q + q [\cos \theta_0 - \delta \sin \theta_0 + O(\delta^2)] \\ &= \alpha_1 - q \sin \theta_0 \frac{1}{aL \sin \theta_0} + O(a^{-2}) \\ &= \alpha_1 - \frac{q}{aL} + O(a^{-2}). \end{aligned} \quad (152)$$

Since $q/a = \beta(1 - q)$ and $L = N/2$,

$$s_1 = \alpha_1 - \frac{2\beta(1 - q)}{N} + O([\beta(1 - q)]^2). \quad (153)$$

Thus

$$1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1 - q)}{N} + O([\beta(1 - q)]^2). \quad (154)$$

For large N ,

$$1 - \alpha_1 = q \left[1 - \cos \frac{\pi}{N} \right] = \frac{q\pi^2}{2N^2} + O(N^{-4}), \quad (155)$$

so

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1 - q)}{N}. \quad (156)$$

7.5 Strong-shortcut endpoint expansion

When a is small, the largest root approaches the zero of U_{L-1} at $\theta = \pi/L$. Write

$$\theta_1 = \frac{\pi}{L} - \varepsilon, \quad 0 < \varepsilon \ll 1. \quad (157)$$

Then

$$\tan(L\theta_1) = \tan(\pi - L\varepsilon) = -L\varepsilon + O(\varepsilon^3), \quad (158)$$

and

$$-a \sin \theta_1 = -a \sin \frac{\pi}{L} + O(a\varepsilon). \quad (159)$$

Thus

$$\varepsilon = \frac{a}{L} \sin \frac{\pi}{L} + O(a^2). \quad (160)$$

Consequently

$$\begin{aligned} s_1 &= 1 - q + q \cos \left(\frac{\pi}{L} - \varepsilon \right) \\ &= \gamma_1 + q\varepsilon \sin \frac{\pi}{L} + O(\varepsilon^2) \\ &= \gamma_1 + \frac{qa}{L} \sin^2 \frac{\pi}{L} + O(a^2). \end{aligned} \quad (161)$$

This shows explicitly how the dominant pole moves from α_1 toward γ_1 as the shortcut strength increases.

8 Consequences for the bimodality discussion

The existence of a second peak is a finite-time shape question. It is not determined solely by which pole is asymptotically largest. In the corrected formula, if

$$F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \quad (162)$$

then local extrema are controlled exactly by the finite difference

$$F_\rho(t+1) - F_\rho(t) = \sum_k B_{\rho k} (s_k - 1) s_k^{t-1}. \quad (163)$$

A second peak at time t_* requires

$$F_\rho(t_*) - F_\rho(t_* - 1) > 0, \quad F_\rho(t_* + 1) - F_\rho(t_*) < 0. \quad (164)$$

These inequalities can be checked exactly from (163). They do not reduce to the original threshold $\beta \leq 2q/[(1-q)N]$.

What can be stated analytically and cleanly is the decay of the right tail of a second peak if it is present:

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad \frac{F_\rho(t+1)}{F_\rho(t)} \rightarrow s_1. \quad (165)$$

Thus the referee-facing corrected decay dependence is

$$\tau_\beta^{-1} = -\log s_1, \quad aT_L \left(1 + \frac{s_1 - 1}{q}\right) + U_{L-1} \left(1 + \frac{s_1 - 1}{q}\right) = 0. \quad (166)$$

For weak shortcuts this gives

$$\tau_\beta^{-1} = -\log \left[\alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2) \right]. \quad (167)$$

Equivalently, in gap form,

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}. \quad (168)$$

The shortcut contribution to the decay gap is the term

$$\frac{2\beta(1-q)}{N}. \quad (169)$$

9 Audit notes: issues found in the original calculation

This section records the mathematical and typographical issues identified during the audit.

9.1 Substantive mathematical issues

1. The shortcut-to-target resolvent sign is wrong for the stochastic interpretation in which a fraction $\beta(1-q)$ of the stay-put probability at u is replaced by a jump to absorbing v . The corrected transient perturbation is negative killing at u , giving (90).
2. The original denominator $1 - z\beta(1-q)\widetilde{W}_u(u, z)$ must be replaced by $1 + z\beta(1-q)\widetilde{W}_u(u, z)$. This changes the pole equation from a positive horizontal-line intersection to a negative one, or equivalently from $a - \phi = 0$ to $a + \phi = 0$.
3. The original first-passage correction has the wrong monotonicity: it decreases the hitting generating function after adding a shortcut to the target. The corrected formula (93) increases it.
4. The original beta condition

$$\beta \leq \frac{2q}{(1-q)N} \quad (170)$$

comes from the wrong-sign denominator and is not a valid corrected condition for the stochastic shortcut-to-target model.

5. The original $t\alpha_1^t$ conclusion comes from an artificial collision between an uncorrected shortcut pole and an absorbing-ring pole. In the corrected model the dominant pole is simple and equals $s_1 < \alpha_1$, so the tail is $B_{\rho 1} s_1^{t-1}$.
6. The reduction of the absorbing-ring double sum contains a sign inconsistency in the intermediate line: with denominator $\cos(\pi j/N) - \cos(2\pi r/N)$, the bracket must be $\gamma_r^t - \lambda_j^t$. The final killed propagator is nevertheless correct after this sign is fixed.
7. The initial-condition check following the killed propagator uses incorrect individual orthogonality claims. The correct cosine and odd-sine sums are (67) and (68); only their combination gives δ_{n,n_0} .
8. In the long convolution formula corresponding to the original Eq. (41), the first no-shortcut term should involve $f_\ell(n_0, v)$, not $f_\ell(n_0, u)$. The baseline term in $F_{n_0}(v, t)$ is the first-passage probability to v .
9. In the same long convolution formula, terms involving the odd-sector propagator $g_r^{(2)}$ should carry α_r , not γ_r . The displayed use of γ_r in the last line is an index/eigenvalue mismatch.
10. The residue coefficient formula for the uncorrected c_k contains bracket mismatches and an apparently incorrect derivative factor. Since that coefficient belongs to the wrong-sign pole problem, it should be replaced rather than repaired.

9.2 Typographical and presentation issues

1. The document class option `idelinks` should be `hidelinks`.
2. The source depends on local style files such as `customcolours.sty`, `customcommands.sty`, and `globalplotconfig.sty`; a portable derivation should avoid these dependencies or include them.
3. A label of the form `\label{eq:}` is not useful and can create reference problems.
4. Several expressions write $G_k^{(1)}(n_0, n, , m)$ and $G_k^{(2)}(n_0, n, , m)$ with a double comma.
5. The text says $Q_{n_0}(n, 0) = \delta_{n,n_0}$ and $Q_{n_0}(n, t) = 0$ in places where it means $W_{n_0}(n, 0)$ and $W_{n_0}(n, t)$.
6. The compact odd-sector sums require N even. This assumption should be stated before using $\sum_{\ell=1}^{N/2}$.
7. The figure caption says “given the requirement $q > 1$ ”; the stochastic model requires $0 < q < 1$, and a nonzero shortcut probability requires $q < 1$ if the shortcut probability is proportional to $1 - q$.
8. The phrase “poles in the dominator” should be “poles in the denominator.”
9. The contour statement after $z = 1/s$ should specify orientation and which singularities are enclosed. The corrected derivation avoids this ambiguity by using partial fractions directly in s .

10 Optional numerical check for the figure parameters

For the original figure parameters

$$N = 10, \quad L = 5, \quad q = \frac{2}{3}, \quad \beta = \frac{2}{7}, \quad (171)$$

we have

$$a = \frac{q}{\beta(1-q)} = 7. \quad (172)$$

The corrected largest root solves

$$\tan(5\theta) = -7 \sin \theta, \quad \frac{\pi}{10} < \theta < \frac{\pi}{5}. \quad (173)$$

Numerically,

$$\theta_1 \approx 0.3865953134, \quad s_1 = 1 - \frac{2}{3} + \frac{2}{3} \cos \theta_1 \approx 0.9507987460. \quad (174)$$

The no-shortcut largest odd eigenvalue is

$$\alpha_1 = 1 - \frac{2}{3} + \frac{2}{3} \cos \frac{\pi}{10} \approx 0.9673710109, \quad (175)$$

so indeed

$$s_1 < \alpha_1. \quad (176)$$

The corrected tail gap is

$$1 - s_1 \approx 0.0492012540. \quad (177)$$